

Xuanyu Zhou

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Research interests: Urban analytics, built environment, urban mobility, data science, Large Language Models

Education

Zhejiang University

B.E. in Urban and Rural Planning, 5 year program 3.93/4.0

Coursework: Principle of Urban Planning, Urban Transportation, Architecture Design, Micro/Macroeconomics

Hangzhou, Zhejiang

2021–2026 (Exp.)

University of California, Berkeley

Exchange in College of Environmental Design

Berkeley, CA

2025

Coursework: Introduction to City Planning, The Structure and Interpretation of Computer Programs, Transit

Research Experiences

MIT-UF-NEU 2025 Joint Summer Research Camp

Remote

Mentee, mentor: Dr. Xuan Jiang

Jun 2025–Oct 2025

- Conducted comprehensive result analysis for a novel GPU-based computational framework for large-scale Urban Air Mobility (UAM) demand modeling and heterogeneous fleet optimization, applied to a case study in the Bay Area.
- Produced spatial and temporal demand visualizations, including geographic heatmaps and fleet-type comparisons, using Python, ArcGIS, Matplotlib, and Plotly.
- Authored the full first draft of the research manuscript as second author, synthesizing findings that demonstrated over 20 minutes of travel time savings for 230,000 selected trips.

Monitoring-Mapping-Modeling (3M) Lab, UC Berkeley

Berkeley, CA

Research Assistant, supervisor: Prof. Lu Liang

Mar 2025–Present

- Designed and implemented a Python-based pipeline to compute urban Sky View Factor (SVF) from street view imagery (SVI) by converting 2D images to 3D point clouds via MoGe, enabling SVF estimation at pedestrian-level positions on sidewalks rather than road-center viewpoints used in conventional SVI approaches.
- Processed over 120,000 street view images across four U.S. cities (San Francisco, Los Angeles, Dallas, Seattle), with planned expansion to major cities nationwide. Conducted cross-source validation comparing SVF derived from SVI, LiDAR, and image-based point clouds.
- Research addresses thermal equity by providing a scalable alternative to coverage-limited LiDAR data, extending accurate SVF and heat exposure estimation to underserved regions including the Global South.

Urban Greenery Equity Research Project

Hangzhou, Zhejiang

Independent Research, advised by Mingze Chen

Jun 2024–Oct 2025

- Independently conducted a longitudinal study on urban green space (UGS) equity in Pittsburgh, a severely shrinking city, analyzing satellite imagery from 2019 to 2022 using machine learning methods to assess changes in UGS distribution, area, and morphology.
- Processed census-tract-level visitation data and applied OLS and Geographically Weighted Regression (GWR) to examine how UGS characteristics influence community vitality in shrinking neighborhoods.
- Authored the full first draft of the manuscript as first author. Presented findings as an oral presentation at the 2024 ACSP Annual Conference. Under review at *Urban Forestry & Urban Greening*.

Qizhenwenxue Undergraduate Research Program, Zhejiang University

Hangzhou, Zhejiang

Research Assistant, supervisor: Prof. Shuang Ma

Aug 2023–Nov 2025

- Authored the Introduction and Literature Review, establishing the theoretical framework on urban mobility and resilience in Nagoya. Contributed to data preprocessing and Geographic Random Forest (GRF) modeling using mobile phone signaling data. Published in *Transactions in Urban Data, Science, and Technology*.
- Independently implemented GRF modeling, including parameter tuning and feature importance analysis, on a dataset of over 600,000 geotagged Weibo posts across four months. Interpreted resulting spatial patterns linking built environment features to resident mental health indicators. Manuscript under review at *Sustainable Cities and Society*.
- Conducted GRF modeling pipeline and data visualization on over 9 million shared bicycle trip records in Shenzhen. Contributed to result interpretation and manuscript conclusion. Under review at *npj Urban Sustainability*.

Publications

Published

[5] Ma, S., **Zhou, X.**, Cai, W., Chen, M., Li, S. (2025). Evaluating the social-economic recovery impacts of built environment for post-pandemic. *Transactions in Urban Data, Science, and Technology*.

Under Review

[4] Jiang, X., **Zhou, X.**, Zhao, Y., Cao, S., Zhao, J., Hansen, M., Sengupta, R. From Patchwork to Network: A Comprehensive Framework for Demand Analysis and Fleet Optimization of Advanced Air Mobility. *IEEE Transactions on Intelligent Transportation Systems*.

[3] **Zhou, X.**, & Chen, M. Greening the Gap: Examining Urban Greenery Equity in Shrinking Cities. *Urban Forestry & Urban Greening*.

[2] Ma, S., Li, W., **Zhou, X.**, Ma, T., He, Y., & Li, S. Future life expectancy extension under bike ride, air pollution exposure, and traffic accidents through modifying the built environment and climate: An integrated modelling study. *npj Urban Sustainability*.

[1] Li, S., Cai, W., Wu, C., Zhang, X., **Zhou, X.**, He, Y., & Ma, S. Exploring the relationship between resident emotions and built environment in Beijing based on a geographical weighted random forest approach. *Sustainable Cities and Society*.

Conference Presentations

[C2] Jiang, X., **Zhou, X.**, Zhao, Y., Cao, S., Zhao, J., Hansen, M., Sengupta, R. (2026). From Patchwork to Network: A Comprehensive Framework for Demand Analysis and Fleet Optimization of Urban Air Mobility. *2026 Transportation Research Board Annual Meeting*. (Accepted)

[C1] **Zhou, X.**, & Chen, M. (2024). Greening the Gap: Examining Urban Greenery Equity in Shrinking Cities. *2024 ACSP Annual Conference*. (Oral Presentation)

Internships

Zhejiang University Urban—Rural Planning & Design Institute Co., Ltd.

Hangzhou, Zhejiang

Assistant Planner

Aug 2025–Nov 2025

- Conducted full life-cycle carbon emission assessment for the Anji Economic Development Zone's national-level Zero-Carbon Industrial Park application, following Chinese national standards.
- Performed GIS-based site selection analysis for low-altitude economy infrastructure across three municipal planning projects; produced spatial visualizations and contributed to stakeholder consultations.

Skills

Programming & Data: Python (pandas, NumPy, geopandas, scikit-learn, Matplotlib, Plotly), R, SQL, Git, $\LaTeX$
Geospatial & Analysis: ArcGIS, QGIS, Google Earth Engine, Machine Learning
Design & Visualization: Rhino, SketchUp, Adobe Suite (Photoshop, Illustrator, InDesign)

Awards

Zhejiang Provincial Government Scholarship , Zhejiang Province (top 3%)	2024, 2025
Second-Prize Scholarship , Zhejiang University (top 10%)	2023, 2024, 2025
Ningbo Future Star Scholarship , Human Resources and Social Security Bureau of Ningbo	2025
Jincheng Academic Scholarship , Zhejiang University	2023
Academic Excellence Award , Zhejiang University	2022, 2023, 2024, 2025
Outstanding Student Leader , Zhejiang University	2022, 2023, 2024

Affiliations

- Student Member**, Association of Collegiate Schools of Planning (ACSP)
- Student Member**, International Association for China Planning (IACP)
- Student Member**, Environmental Design Research Association (EDRA)